10-planarity-coloring：平面图与图着色 (Planarity and Coloring)

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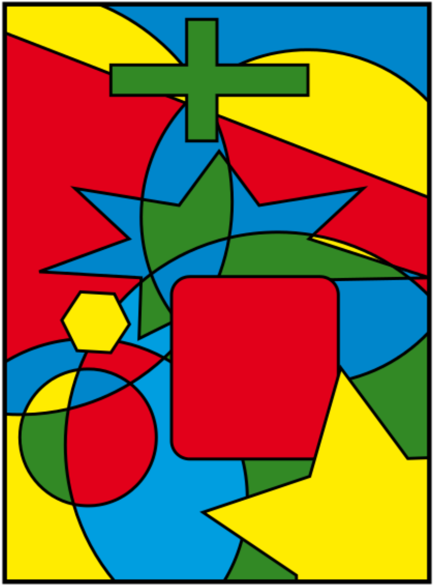
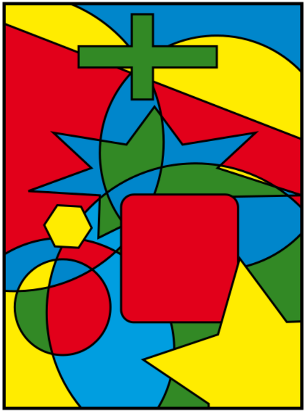
2026 年 05 月 21 日



Theorem (Four Color (Map) Theorem (informal))

Every map can be colored with only four colors such that no two adjacent regions share the same color.



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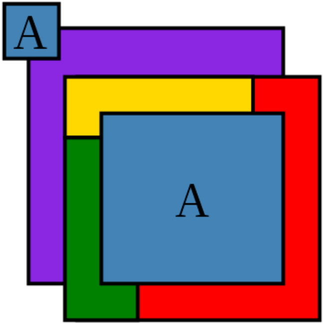
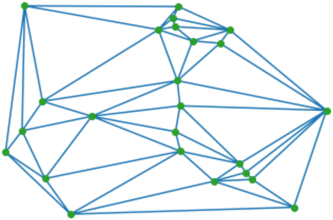
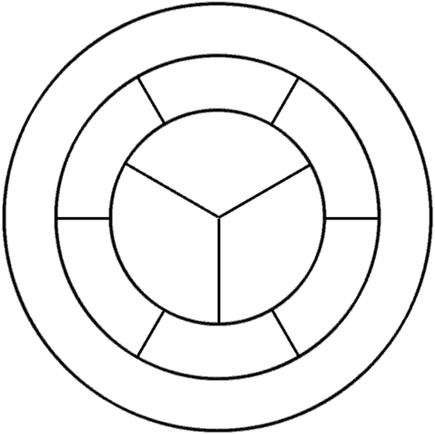


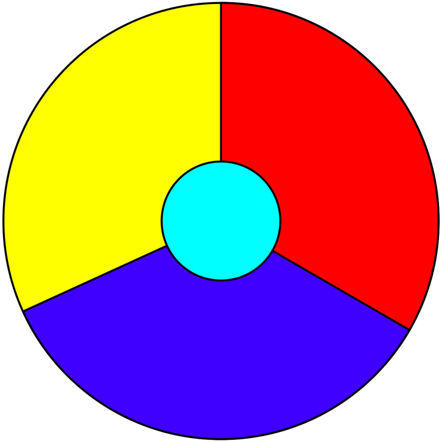
Regions should be contiguous.



Adjacent regions share a segment.



4 colors are necessary

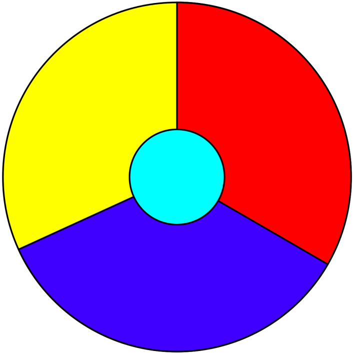


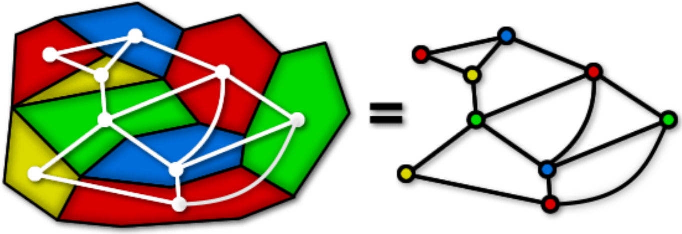
Every region is adjacent to the other 3 regions.

What if we have a map which contains 5 regions so that every region is adjacent to the other 4 regions?

IMPOSSIBLE™

What does Four Color Theorem to do with Graph Theory?





Every map produces a planar graph.

Theorem (Four Color Theorem (Kenneth Appel, Wolfgang Haken; 1976))

Every simple planar graph is 4-colorable.

I will *not* show its proof (which I don't understand either)!

Theorem (Four Color Theorem (Kenneth Appel, Wolfgang Haken; 1976))

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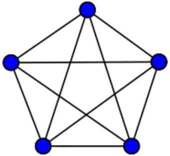


What if we have a map which contains 5 regions so that every region is adjacent to the other 4 regions?

Theorem

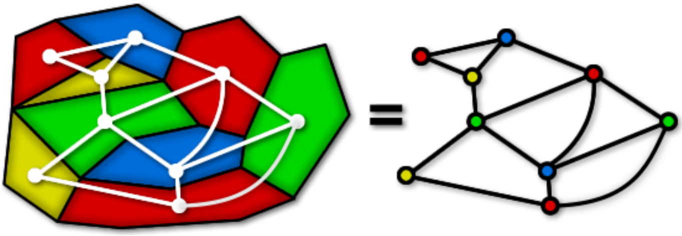
K_5 *is not planar.*

Theorem

Every simple planar graph is 6-colorable.

Theorem (Percy John Heawood (1890))

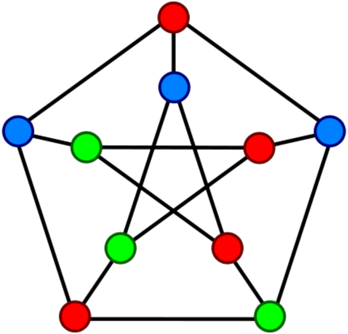
Every simple planar graph is 5-colorable.



Graph Coloring Problem

Definition (k-Colorable (k-可着色的))

If G is a connected undirected graph without loops,
then G is k-colorable if its vertices can be colored in k colors
so that adjacent vertices have different colors.

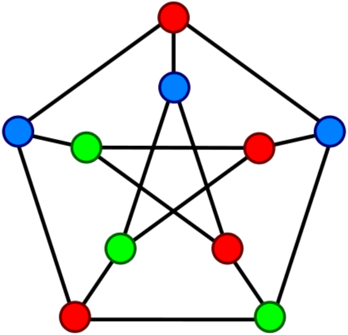


The Petersen graph is ≥ 3-colorable.

Definition (k-Chromatic (k-色数的))

If G is k-colorable, but is not $(k-1)$-colorable, then G is k-chromatic.

$$\chi(G) = k$$



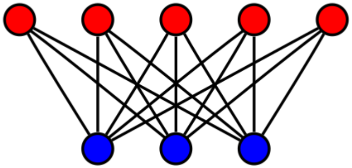
The Petersen graph is 3-chromatic.
(It contains an odd cycle.)

Lemma

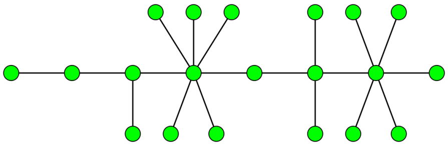
A graph is 1*-colorable iff it is the the empty graph with 1 vertex.*

Theorem

A graph is 2-colorable iff it is bipartite.



$K_{5,3}$

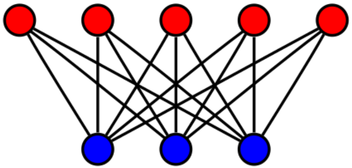


Trees are bipartite.

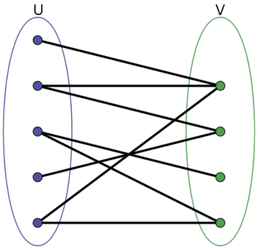
Theorem (Characterization of Bipartite Graphs)

A graph is 2-colorable iff it is bipartite.

A graph is bipartite iff it does not contain any odd cycles.



$K_{5,3}$

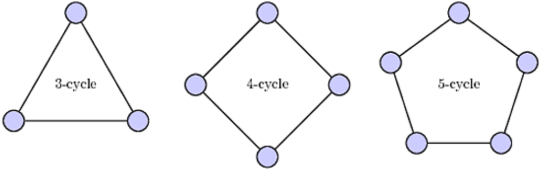


Lemma

Every tree is bipartite and is thus 2-colorable.

Lemma (Characterization of Bipartite Graphs ($\Longrightarrow$))

If a graph is bipartite, then it does not contain any odd cycles.



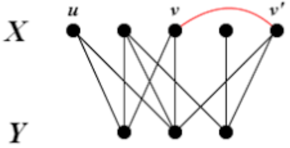
Lemma (Characterization of Bipartite Graphs ($\Longleftarrow$))

If a graph G does not contain any odd cycles, then it is bipartite.

Choose any vertex $u \in V(G)$.

$$X = \{x \in V(G) : \text{the distance from } u \text{ to } x \text{ is even}\}$$

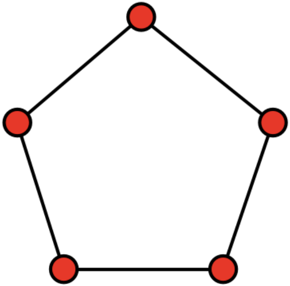
$$Y = \{y \in V(G) : \text{the distance from } u \text{ to } y \text{ is odd}\}$$

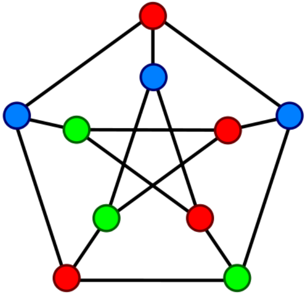


$u \rightsquigarrow v \to v' \rightsquigarrow u$ is an odd cycle.

Theorem

The 3-coloring problem (i.e., testing whether a graph is 3-colorable or not) is NP-complete (HARD!).





Theorem

The 4-coloring problem is NP-complete (HARD!).

Theorem (Four Color Theorem (Kenneth Appel, Wolfgang Haken; 1976))

Every simple planar graph is 4-colorable.

Theorem

Let G be a simple connected graph. Then,

$$\chi(G) \leq \Delta(G) + 1.$$

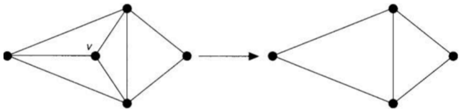
By induction on the number of vertices of G.

Basis Step: $n = 1$. $\chi(G) = 1$ and $\Delta(G) = 0$.

Induction Hypothesis: Suppose that for any simple connected graph G with n vertices,

$$\chi(G) \leq \Delta(G) + 1.$$

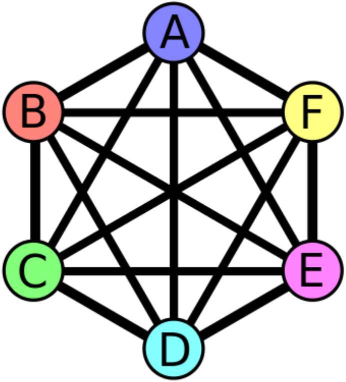
Induction Step: Consider a simple connected graph G with $n + 1$ vertices. $\deg(v) \leq \Delta(G)$

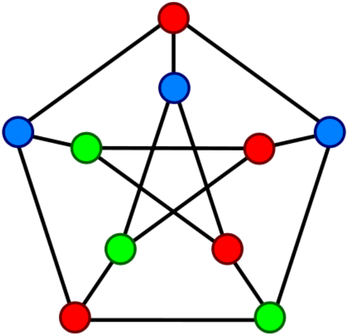


Theorem (Brooks's Theorem (R. Leonard Brooks; 1941))

Let G be a simple connected graph other than a complete graph or an odd cycle. Then

$$\chi(G) \leq \Delta(G).$$





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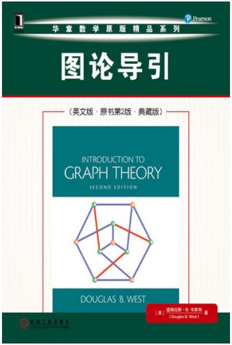
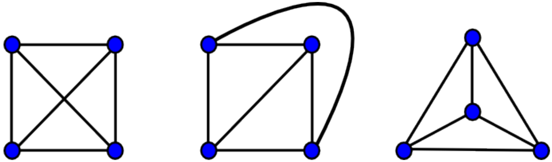


Theorem 5.1.22

Definition (Planar Graph (平面图))

A planar graph is a graph that can be drawn in the plane without edge crossings.

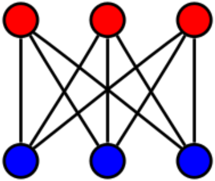


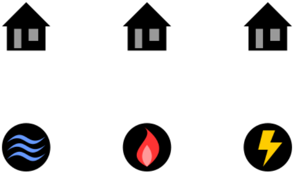
Theorem (K. Wagner (1936); I. Fáry (1948))

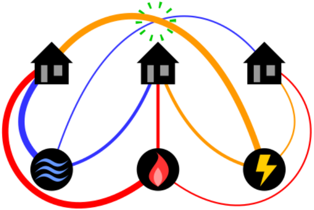
Every simple planar graph can be drawn with straight lines.

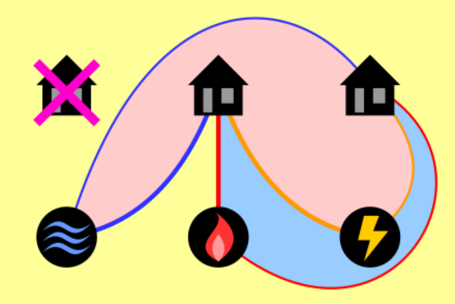
Theorem (Kazimierz Kuratowski, 1930)

The utility graph $K_{3,3}$ is non-planar.

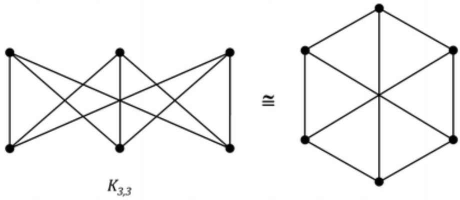


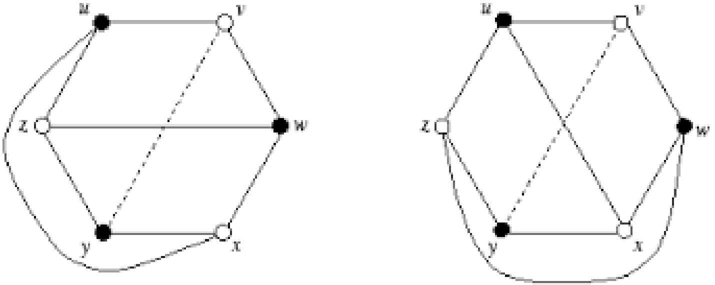






Proof without Words

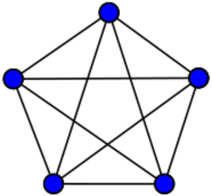


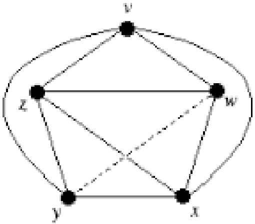


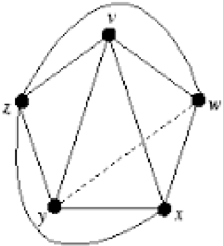
$$\textcolor{red}{\mathrm{cr}}(K_{3,3}) = 1 \quad \text{(crossing number)}$$

Theorem

K_5 is non-planar.







$$\color{red}{\mathrm{cr}}(K_5) = 1$$

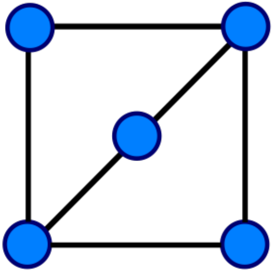
Theorem (Kazimierz Kuratowski, 1930)

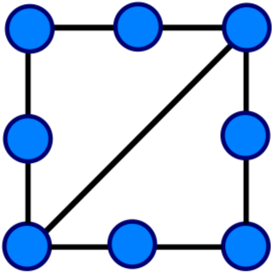
A graph is planar iff it contains no subgraph homeomorphic to K_5 or $K_{3,3}$.

"The K in K_5 stands for Kazimierz,
and the K in $K_{3,3}$ stands for Kuratowski."

Definition (Homeomorphic)

Two graphs are homeomorphic if one can be obtained from another by inserting or contracting vertices of degree 2.





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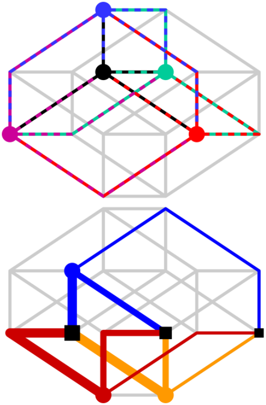
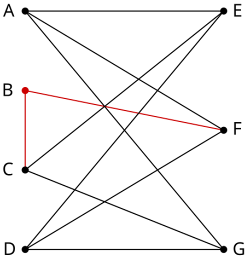
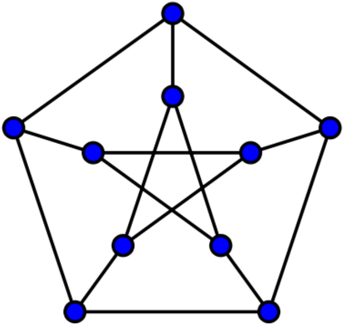
A graph is planar iff it contains no subgraph homeomorphic to K_5 or $K_{3,3}$.



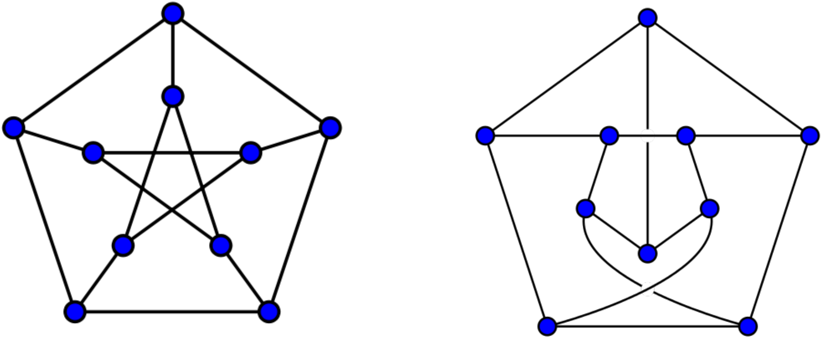


Theorem

The Petersen graph is non-planar.



https://github.com/courses-at-hnu-by-hfwei/math4cs-lectures/blob/main/10-planarity-coloring/figs/Kuratowski-Petersen.gif



$\textcolor{red}{\mathrm{cr}}(\text{Petersen Graph}) = 2$

Efficient Planarity Testing

JOHN HOPCROFT AND ROBERT TARJAN

Cornell University, Ithaca, New York

ABSTRACT. This paper describes an efficient algorithm to determine whether an arbitrary graph G can be embedded in the plane. The algorithm may be viewed as an iterative version of a method originally proposed by Auslander and Parter and correctly formulated by Goldstein. The algorithm uses depth-first search and has $O(V)$ time and space bounds, where V is the number of vertices in G. An ALGOL implementation of the algorithm successfully tested graphs with as many as 900 vertices in less than 12 seconds.

KEY WORDS AND PHRASES: algorithm, complexity, depth-first search, embedding, genus, graph, palm tree, planarity, spanning tree

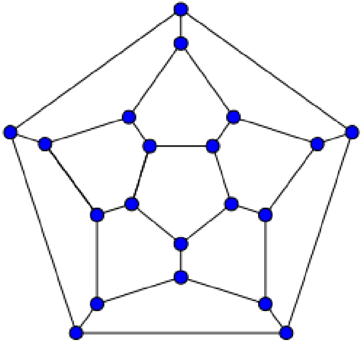
CR CATEGORIES: 3.24, 5.25, 5.32

A planar graph should not has too many edges.

Theorem (Euler's Formula, 1750)

Let G be a plane drawing of a connected planar graph, and let n, m, and f denote respectively the number of vertices, edges, and faces of G.

$$n - m + f = 2$$



$$n - m + f = 20 - 30 + 12 = 2$$

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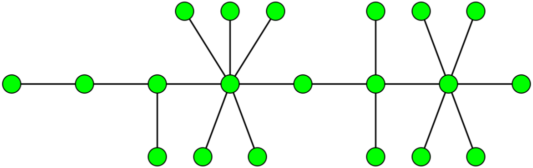


$$n - m + f = n - (n-1) + 1 = 2$$

By induction on the number of edges of G.

Basis Step: $m = 0$. We have $n = 1$ and $f = 1$.

Induction Hypothesis: It holds for plane graphs with m edges.

Induction Step: Consider a plane graph G with $m + 1$ edges.

If G is a tree, we are done.

Otherwise, G contains a cycle.

Let e be an edge in some cycle of G.

Consider $G' = G - e$.

$$n - (m - 1) + (f - 1) = 2$$

Therefore,

$$n - m + f = 2$$

Theorem

Let G be a simple connected planar graph with $n \geq 3$ vertices and m edges. Then

$$m \leq 3n - 6.$$

$$n - m + f = 2$$

$$2m = \Sigma_f \deg(f) \geq 3f$$

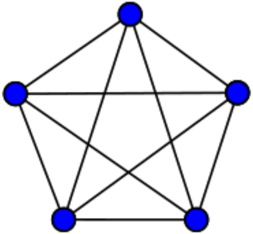
Double Counting the Edges around Faces:

each edge bounds 2 faces;

each face is bounded by ≥ 3 edges

Theorem

K_5 is non-planar.

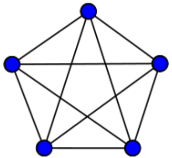


$$m \leq 3n - 6$$

$$10 \leq 3 \times 5 - 6$$

Theorem

K_5 is non-planar.



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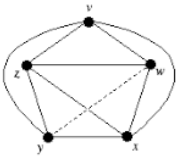
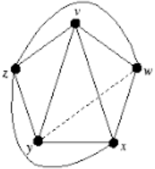
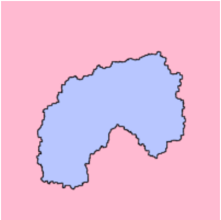




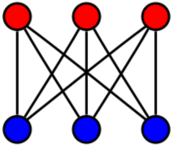
Theorem (Jordan Curve Theorem (1887))



Algebraic Topology

Theorem

$K_{3,3}$ is non-planar.



$$m \leq 3n - 6$$

$$9 \leq 3 \times 6 - 6$$

Theorem

Let G be a simple connected planar graph with $n \geq 3$ vertices and m edges. If G has no triangles, then

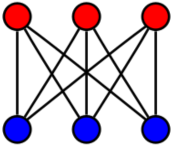
$$m \leq 2n - 4.$$

$$n - m + f = 2$$

$$2m = \Sigma_f \deg(f) \geq 4f$$

Theorem

$K_{3,3}$ is non-planar.



$$m \leq 2n - 4$$

$$9 \leq 2 \times 6 - 4$$

Theorem

Every simple planar graph contains a vertex of degree ≤ 5.

Suppose that, by contradiction, $\delta(G) \geq 6$.

$$2m = \Sigma_v \deg(v) \geq 6n$$

$$m \leq 3n - 6$$

$$3n \leq m \leq 3n - 6$$

Theorem

Every simple planar graph is 6-colorable.

By induction on the number of vertices.

Basis Step: $n = 1$. Trivial.

Induction Hypothesis: Suppose that it holds for simple planar graphs with $n \geq 1$ vertices.

Induction Step: Consider a simple planar graph G with $n + 1$ vertices.

G contains a vertex v of degree ≤ 5.

$G' = G - v$ is 6-colorable.

Thus, G is 6-colorable. $(\deg(v) \leq 5)$

Theorem (Percy John Heawood (1890))

Every simple planar graph is 5-colorable.

By induction on the number of vertices.

Basis Step: $n = 1$. Trivial.

Induction Hypothesis: Suppose that it holds for simple planar graphs with $n \geq 1$ vertices.

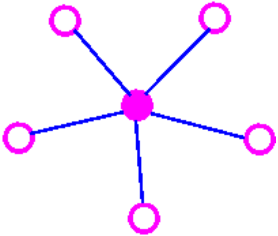
Induction Step: Consider a simple planar graph G with $n + 1$ vertices.

G contains a vertex v of degree ≤ 5.

$G' = G - v$ is 5-colorable.

If $\deg(v) < 5$, G is 5-colorable.

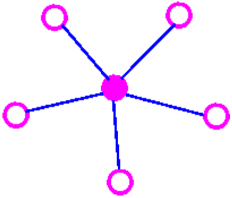
Now assume that $\deg(v) = 5$.



$$\{v, v_1\}, \{v, v_2\}, \{v, v_3\}, \{v, v_4\}, \{v, v_5\}$$

If v_1, v_2, v_3, v_4, and v_5 uses < 5 colors, we are done.

Now assume that v_1, v_2, v_3, v_4, and v_5 uses 5 colors.



Suppose that there is *no* $v_1 \sim v_3$ path in $G' = G - v$, all of whose vertices are colored red or blue.

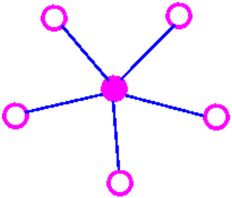
Let S be the set of all red or blue vertices of $G - v$ connected to v_1 by a red-blue path.

$$v_1 \in S, \quad v_3 \notin S$$

Interchange the colors of the red or blue vertices in S

Coloring v red produces a 5-coloring of G.

Now assume that there *is* a $v_1 \sim v_3$ path in $G' = G - v$,
all of whose vertices are colored red or blue.



There cannot be $v_2 \sim v_4$ path in $G' = G - v$,
all of whose vertices are colored green or purple.

(Otherwise, G' and thus G is non-planar.)

By similar argument, G is 5-colorable.

$P(G, k)$: How many ways to color the vertices of G with k colors?

$$P(P_4, k) = k(k-1)^3$$

$$P(C_4, k) = k(k-1)^2(k-2)$$

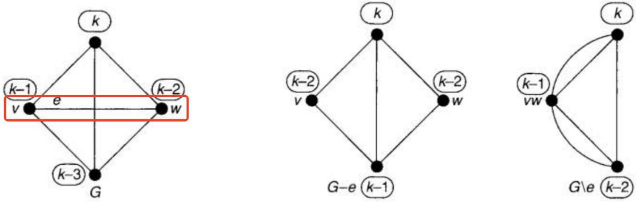
$$P(K_4, k) = k(k-1)(k-2)(k-3)$$

Theorem (Deletion-Contraction Recurrence)

Let G be a simple graph and e be an edge of G. Then

$$P(G, k) = P(G - e, k) - P(G/e, k),$$

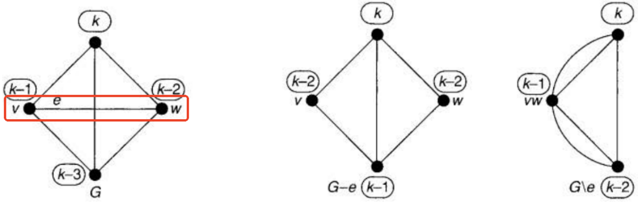
where $G - e$ is the graph obtained from G by deleting e, and G/e is the graph obtained from G by contracting e.



$$k(k-1)(k-2)(k-3) = k(k-1)(k-2)^2 - k(k-1)(k-2)$$

Theorem (Deletion-Contraction Recurrence)

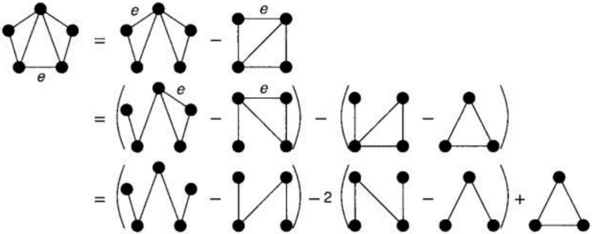
$$P(G,k) = P(G-e,k) - P(G/e,k)$$



$$P(G-e,k) = P(G,k) + P(G/e,k)$$

$$\begin{aligned} P(G-e,k) &= P(G-e,k) \text{ with } v \text{ and } w \text{ having } \textit{different} \text{ colors} \\ &+ P(G-e,k) \text{ with } v \text{ and } w \text{ having the } \textit{same} \text{ color} \end{aligned}$$

$$P(G,k) = P(G-e,k) - P(G/e,k)$$



$$\begin{aligned} P(G,k) &= k(k-1)^4 - 3k(k-1)^3 + 2k(k-1)^2 + k(k-1)(k-2) \\ &= k^5 - 7k^4 + 18k^3 - 20k^2 + 8k \end{aligned}$$

Theorem (Chromatic Polynomial)

$P(G,k)$ is a polynomial in k.

$$k^n - mk^{n-1} + \cdots - \cdots + ak + 0$$


Thank
You!

Your opinon
Matters

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